

# AI Agent Engineering

## From Fundamentals to Production — A 12-Week Certificate Program

|                               |                                 |                                           |                               |
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| Duration: 12 Weeks (3 Months) | Total Volume: 132 Contact Hours | Format: Theory + Hands-on Labs + Capstone | Level: Production Engineering |
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### Program Overview

This program takes students from the fundamentals of AI agents to designing, building, evaluating, and deploying a production-grade multi-agent system. Rather than a set of disconnected tutorials, the course follows a **single progressive build**: each week adds a new capability to a project the student carries forward, so that by the end of the program every student has a working, evaluated, portfolio-ready agent application.

**The Industry Learning Cycle:** Every topic follows a rigorous 7-stage learning cycle: **Concept** → **Architecture** → **Demonstration** → **Build** → **Break** → **Debug** → **Evaluate**. This ensures students don't just understand agents conceptually — they can build, test, and fix them the way it is done in production engineering.

### Program Structure & Curriculum Breakdown

| Component            | Hours | Purpose & Focus                                                      |
|----------------------|-------|----------------------------------------------------------------------|
| Theory / Concepts    | 66 h  | Understand how agent architectures and reasoning loops actually work |
| Hands-on Labs        | 44 h  | Build and test individual modular components in sandboxed runtimes   |
| Projects / Capstone  | 18 h  | Integrate everything into one comprehensive production system        |
| Evaluation / Defense | 4 h   | Benchmark, test against adversarial attacks, and defend the system   |

### Technical Prerequisites

| Required Competencies                                                                                                                                                                                                       | Not Required (We Teach These)                                                                                                                                                                                                    |
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| <ul style="list-style-type: none"><li>• Basic Python programming &amp; logic</li><li>• Git fundamentals &amp; GitHub basics</li><li>• APIs and JSON data structures</li><li>• Basic command line / terminal usage</li></ul> | <ul style="list-style-type: none"><li>• Advanced mathematics or calculus</li><li>• Deep learning research / paper reproduction</li><li>• Transformer architecture internals</li><li>• Reinforcement learning expertise</li></ul> |

\* Note: Students without a programming background can take an optional 8–10 hour Python, API, and Git bridge module before the program begins.

### Week-by-Week Curriculum (Weeks 1 to 12)

| Week   | Module & Title         | Key Concepts                                                                                                               | Hands-on Practical Work                                                                                                 |
|--------|------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Week 1 | What is an AI Agent?   | Agent loop, LLM vs Agent, ReAct (Think → Act → Observe), workflow vs agent, agent architectures, when not to use an agent. | Build progressively from a simple LLM app to a tool-using ReAct agent. Mini Project: your first functional AI agent.    |
| Week 2 | Context Engineering I  | Messages, system/user/tool roles, context window, tokenization, prompt architecture, static vs. dynamic context.           | Build a context-aware agent and a context builder pipeline; observe how context changes model output.                   |
| Week 3 | Context Engineering II | Prompt engineering, tool descriptions, prompt injection, defensive prompting, context compression, long conversations.     | Project: a production-style customer support agent with business rules and injection defenses — then attack and fix it. |
| Week 4 | Memory Engineering     | Short-term vs. long-term memory, episodic vs. semantic memory, memory extraction, retrieval, lifecycle, and conflicts.     | Build a personal AI assistant with persistent memory across conversations.                                              |

|         |                                    |                                                                                                                         |                                                                                                                       |
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| Week 5  | RAG Engineering                    | Chunking, embeddings, vector databases, hybrid search, reranking, metadata filtering, retrieval failure modes.          | Build a full retrieval-augmented generation pipeline and compare chunking and retrieval strategies.                   |
| Week 6  | Tool Engineering                   | Tool schemas, function calling, tool selection and reliability, error handling, permissions, and security.              | Build multiple tools (search, calculator, database, API) and a tool-using research agent.                             |
| Week 7  | MCP & Production Tooling           | MCP concepts, servers and clients, resources, authentication, permissions, and tool discovery.                          | Connect and register multiple MCP tools with permission controls. Milestone: a personal AI work agent.                |
| Week 8  | Coding Agents                      | Coding agent architecture, repository understanding, code generation, sandboxing, testing, and the agentic coding loop. | Project: a coding agent that inspects a real repository, edits files, runs tests, and fixes errors.                   |
| Week 9  | Async, Event-Driven & Computer Use | Synchronous vs. asynchronous agents, event-driven architecture, human-in-the-loop, browser automation basics.           | Build an event-driven agent triggered by an incoming event, plus a controlled browser-automation mini-lab.            |
| Week 10 | Agent Evaluation                   | Why demos aren't evaluation, benchmarks, trajectory evaluation, LLM-as-judge, reliability, latency, cost, safety.       | Build an evaluation framework for your own agent and use it to find and log failure patterns.                         |
| Week 11 | Agent Improvement                  | When to improve the prompt, context, tools, or the model itself; version comparison; continuous evolution loop.         | Take your Week 10 agent, diagnose failures, and ship an improved Version 2 — then compare results.                    |
| Week 12 | Multi-Agent Systems & Capstone     | Manager-worker and peer-to-peer patterns, shared vs. independent context, delegation, coordination, failure handling.   | Capstone: a multi-agent research system (manager, researcher, analyst, critic, synthesizer) — presented and defended. |

## Capstone Project — Autonomous Multi-Agent Swarm

The final project is not a simple chatbot. Every student builds a complete, production-grade agent system that synthesizes every capability learned across the 12 weeks into a single, cohesive architecture:

- **An Agent Router:** Directs complex, multi-modal requests dynamically to specialized sub-agents and tool execution planes.
- **Context Engine & Memory System:** Maintains grounded, stateful persona context with persistent episodic and semantic recall.
- **Tools & MCP Integrations:** Executes real-world operations via Model Context Protocol servers and secure API dispatchers.
- **RAG & Coding Agent Pipeline:** Hybrid document search with AST-aware code editing, sandboxed execution, and automated test fixes.
- **Evaluation & Telemetry Framework:** Live benchmark logging of accuracy, token economics, latency, and guardrail violation rates.
- **Continuous Evolution Cycle:** Fully documented diagnostic and improvement cycle transitioning the agent from Version 1 to Version 2.

**Portfolio Deliverable:** Students graduate with a production GitHub repository featuring architectural diagrams, evaluation traces (Arize Phoenix / LangSmith), sandboxed integration tests, and a defensible multi-agent deployment.

## Graduate Learning Outcomes

By the conclusion of this program, students will be verified capable to:

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| <ul style="list-style-type: none"><li>✓ Design an end-to-end production AI agent architecture</li><li>✓ Work with LLM APIs and engineer dynamic context pipelines</li><li>✓ Build memory systems and hybrid RAG search indices</li><li>✓ Connect external tools and Anthropic MCP servers</li><li>✓ Build autonomous coding agents that inspect, test, and repair repos</li></ul> | <ul style="list-style-type: none"><li>✓ Implement asynchronous, event-driven agent workflows</li><li>✓ Evaluate agent performance and debug trajectory failures</li><li>✓ Optimize multi-agent swarms for cost, latency, and reliability</li><li>✓ Apply agent security principles including prompt-injection defense</li><li>✓ Deploy, monitor, and continuously improve agents in production</li></ul> |
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